

# THE ZERO PARADOX

## ZP-L: Incomputability Convergence

Initial: May 2026 | Current: 1.2, June 2026

All theorems §I–§VII proved sorry-free in Lean 4. Axiom footprint: [propext, Classical.choice, Quot.sound] throughout.

ZP-L establishes four results connecting the formal axioms of the ZP framework to standard results in ordinal theory and computability. First, Classical.choice appears at the non-constructive diagonal step in each of the four mathematical settings of the ZP framework — topology, information theory, set theory, and computation. Second, Roger’s fixed-point theorem (Kleene’s second recursion theorem) is formalized as a wrapper, formalizing the computational fixed-point structure. Third, the ordinal  $\epsilon_0$  is fully characterized as the first fixed point of  $\alpha \mapsto \omega^\alpha$  and the limit of the tower  $\omega, \omega^\omega, \omega^{\omega^\omega}, \dots$ . Fourth, ordinals below  $\epsilon_0$  encode into  $\mathbb{Z}_2$  via their Cantor normal form, and as the tower stages approach  $\epsilon_0$ , their 2-adic encodings converge to  $0 = \perp$ .

The central result (§VII) is the canonical snap map:  $\phi \alpha = \text{if } \alpha < \epsilon_0 \text{ then } c_0 \text{ else } c_1$  simultaneously satisfies all five conditions — monotone, tower-aligned, fixed-point-respecting, snapping at  $\epsilon_0$ , and  $\epsilon_0$  minimal. All conditions are verified without free hypotheses for this witness. 24 theorems proved, zero sorry, axiom footprint [propext, Classical.choice, Quot.sound] throughout.

### Section I: Axiom Footprint Convergence

Non-constructibility appears in four mathematical settings across the ZP framework. Classical.choice is required at each diagonal step in these proofs — a constructive alternative was not found in any of the four settings.

| Layer | Formal Language          | Expression of non-constructibility                      |
|-------|--------------------------|---------------------------------------------------------|
| ZPB   | Topology                 | C3: no continuous path $\perp \rightarrow x \neq \perp$ |
| ZPC   | Information Theory       | L-INF: infinite surprisal at $\perp$                    |
| ZPJ/K | Set Theory + Computation | bot_self_mem (AFA); botCode (Kleene)                    |
| ZPI   | Algorithmic IT           | $K(S_n n)/ S_n  \rightarrow 1$ ; K uncomputable         |

#### Remark: Why K is Absent from Lean

Kolmogorov complexity K is not computed in Lean in this framework. Its existence as a total function requires Classical.choice — exactly the axiom Nat.Partrec.Code.fixed\_point<sub>2</sub> already uses in ZP-K. The AFA/Kleene route reaches the same fixed-point structure via a provable path without requiring K to be explicitly computed.

Axiom footprint evidence — the following ZP-K theorems all carry [propext, Classical.choice, Quot.sound]:

### Remark: Why K is Absent from Lean

ZPK.t\_comp (T-COMP four-way equivalence)

ZPK.da1\_paths\_unified

ZPK.isComputationalQuine\_undecidable

ZPK.infinite\_quine\_family

The Classical.choice entry is the computational expression of the diagonal. ZP-L inherits this footprint throughout.

## Section II: Roger Fixed-Point Stability

Roger's fixed-point theorem (also known as Kleene's second recursion theorem) states that any computable transformation of a code has a behavioral fixed point: a code  $c$  such that running  $f(c)$  and running  $c$  produce the same partial function. For any computable transformation, at least one fixed-point code exists.

### Theorem: roger\_fixed\_point\_stability (ZPL.lean §II)

For any computable  $f : \text{Code} \rightarrow \text{Code}$ ,

$\exists c : \text{Code}, \text{eval } (f\ c) = \text{eval } c$

Proof: wrapper around ZPK.roger\_fixed\_point\_exists.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Note: this is an existential result. The specific code  $c$  is not constructively produced; Classical.choice selects it. This is the same non-constructive step that appears across all ZP layers (§I).

## Section III: The Ordinal $\varepsilon_0$

The ordinal  $\varepsilon_0$  is the smallest fixed point of the map  $\alpha \mapsto \omega^\alpha$ . It is the supremum of the tower  $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$  — each stage is strictly below  $\varepsilon_0$ , and  $\varepsilon_0$  is not reached by any finite iteration. This section is entirely within Lean scope via Mathlib's ordinal machinery.

### Definitions (ZPL.lean §III)

$\text{epsilonZero} : \text{Ordinal} := \text{Ordinal.epsilon } 0 (= \text{nfp } (\omega^\cdot) 0 = \text{veblen } 1\ 0)$

$\text{fundamentalSeq} : \mathbb{N} \rightarrow \text{Ordinal} := \text{fun } n \mapsto (\alpha \mapsto \omega^\alpha)^\wedge[n]\ 0$

Explicit stages:

$\text{fundamentalSeq } 0 = 0$

$\text{fundamentalSeq } 1 = 1$

### Definitions (ZPL.lean §III)

$\text{fundamentalSeq } 2 = \omega$

$\text{fundamentalSeq } 3 = \omega^\omega$

$\text{fundamentalSeq } (n+1) = \omega^{(\text{fundamentalSeq } n)}$

### Theorem: epsilonZero\_fixedPoint

$\omega^{\varepsilon_0} = \varepsilon_0$

Proof: Ordinal.omega0\_opow\_epsilon 0 (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: epsilonZero\_eq\_nfp

$\varepsilon_0 = \text{nfp } (\omega^{\cdot}) 0$

Proof: Ordinal.epsilon\_zero\_eq\_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: epsilonZero\_eq\_iSup

$\varepsilon_0 = \sqcup n : \mathbb{N}, \text{fundamentalSeq } n$

Proof: iSup\_iterate\_eq\_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: epsilonZero\_tower\_lt

$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

Every finite stage of the tower is strictly below  $\varepsilon_0$ .

Proof: Ordinal.iterate\_omega0\_opow\_lt\_epsilon\_zero (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: epsilonZero\_le\_fixedPoint

$\forall b : \text{Ordinal}, \omega^b = b \rightarrow \varepsilon_0 \leq b$

$\varepsilon_0$  is the least fixed point of  $\omega^{\cdot}$  above 0.

Proof: Ordinal.epsilon\_zero\_le\_of\_omega0\_opow\_le (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: fundamentalSeq\_strictMono

$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \text{fundamentalSeq } (n + 1)$

The tower is strictly monotone.

Proof: by induction; successor step uses `isNormal_opow`.

Lean purity: `[propext, Classical.choice, Quot.sound]`. ✓

### Remark R-L.1: Proof-Theoretic Alignment

Gentzen's theorem (1936) establishes that  $\epsilon_0$  is the proof-theoretic ordinal of Peano Arithmetic: PA can prove transfinite induction for any ordinal strictly below  $\epsilon_0$ , but not for  $\epsilon_0$  itself. This is not claimed or proved here.

ZP-L derives  $\epsilon_0$  as the snap threshold from ordinal fixed-point structure, independently of proof theory. Both derivations locate the same boundary: the ordinal where  $\omega$ -tower self-iteration becomes self-limiting. No claim is made that this alignment is more than a structural observation.

### Remark R-L.2: Surreal Numbers

Every ordinal is a surreal number (Conway, 1976), so  $\epsilon_0$  lives in the surreal number field  $\text{No}$ .  $\text{No}$  satisfies the axioms of a real-closed field, and by Tarski's completeness theorem for the theory of real-closed fields, every first-order sentence in the language of ordered fields that holds in  $\mathbb{R}$  also holds in  $\text{No}$ , and vice versa.

ZP-F (The Counterexamples) proves that the Binary Snap — the forced transition from  $\perp$  to the first non-null ordinal threshold ( $\epsilon_0$ , established in §V–§VII above) — cannot occur in any linearly ordered field. This result applies directly to  $\text{No}$  considered as a linearly ordered field.

The surreals therefore contain  $\epsilon_0$  as an ordinal while simultaneously satisfying the density condition that blocks the Binary Snap in their ordered field structure. Both structures coexist in  $\text{No}$ :  $\epsilon_0$  is present as an ordinal, and the field density that blocks the snap is present in the field structure. The two results apply to different structural aspects of  $\text{No}$ .

### Remark R-L.3: Hyperreals and Łoś's Theorem

The hyperreals  ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/U$  (ultrafilter  $U$  on  $\mathbb{N}$ ) satisfy a transfer result: by Łoś's theorem, every first-order sentence true in  $\mathbb{R}$  holds in  ${}^*\mathbb{R}$ . Łoś's theorem applies to the full first-order theory, so any first-order property of  $\mathbb{R}$  — including field density — transfers.

Field density transfers: between any two hyperreals there is another. The density condition that blocks the Binary Snap in  $\mathbb{R}$  (proved in ZP-F) therefore holds in  ${}^*\mathbb{R}$  as well.

The non-standard naturals  ${}^*\mathbb{N}$  contain infinite elements, but every infinite  $H \in {}^*\mathbb{N}$  has a predecessor  $H-1$  in  ${}^*\mathbb{N}$ .  $\epsilon_0$  is a limit ordinal: it has no predecessor and is the supremum of the tower below it.  ${}^*\mathbb{N}$  cannot host the Binary Snap not only because of field density, but because its infinite elements are successor-like rather than limit-like. The snap requires genuine limit ordinal structure.

The monad of 0 in  ${}^*\mathbb{R}$  — the external set  $\{\epsilon : |\epsilon| < 1/n \text{ for all standard } n\}$  — is external to the ultrapower. Between any two distinct elements of the monad there is another (their arithmetic mean in  ${}^*\mathbb{R}$ ), so no discrete jump at 0 is possible in  ${}^*\mathbb{R}$ .

### Remark R-L.3: Hyperreals and Łoś's Theorem

Field density is an internal property of the ordered field structure of  ${}^*\mathbb{R}$  and transfers by Łoś's theorem. Limit ordinal structure is set-theoretic and is not preserved by the ultrapower construction:  ${}^*\mathbb{N}$  contains only successor-like infinite elements, not limit-like ones. The two results — transfer of density, failure of limit-ordinal structure — come from different theorems.

## Section IV: Cantor Normal Form Bridge

Every ordinal below  $\varepsilon_0$  has a unique Cantor normal form — a finite sum  $a_1 \cdot \omega^{e_1} + a_2 \cdot \omega^{e_2} + \dots$  with strictly decreasing exponents  $e_1 > e_2 > \dots$  and each coefficient a nonzero natural number (a nonzero ordinal strictly below  $\omega$ ). In Lean: `NONote` (`Mathlib.SetTheory.Ordinal.Notation`). The encoding `cnfToZp2` maps each such ordinal to  $\mathbb{Z}_2$  via structural recursion on the Cantor normal form.

### Definition: `cnfToZp2` (ZPL.lean §IV)

`cnfToZp2` : `NONote`  $\rightarrow$   $\mathbb{Z}_2$

Base: `cnfToZp2(0)` = 0

Recursive: `cnfToZp2( $\omega^e \cdot n + a$ )` =  $2^{(v_2(\text{cnfToZp2}(e)) + 1)} \cdot n + \text{cnfToZp2}(a)$  [ $n : \mathbb{N}^+$ ]

where  $v_2$  denotes the 2-adic valuation.

Valuation of the  $n$ -th tower stage:

`cnfToZp2(towerNONote 0)` = 0 (Lean: `PadicInt.valuation 0` = 0; standard:  $v_2(0) = +\infty$ )

`cnfToZp2(towerNONote 1)` = 2 (valuation 1)

`cnfToZp2(towerNONote 2)` = 4 (valuation 2)

`cnfToZp2(towerNONote n)` =  $2^n$  (valuation  $n$ )

### Theorem: `cnfToZp2_tower_valuation`

$\forall n : \mathbb{N}, (\text{cnfToZp2 } (\text{towerNONote } n)).\text{valuation} = n$

The 2-adic valuation of the  $n$ -th tower stage encoding equals  $n$ .

Proof: by induction; successor step uses `PadicInt.valuation_pow`.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

### Theorem: `cnfToZp2_valuation_unbounded`

$\forall k : \mathbb{N}, \exists \alpha : \text{NONote}, k \leq (\text{cnfToZp2 } \alpha).\text{valuation}$

The 2-adic valuation is unbounded across ordinals below  $\varepsilon_0$ .

### Theorem: cnfToZp2\_valuation\_unbounded

Proof: towerNONote k witnesses the bound.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: tower\_converges\_to\_zero

Filter.Tendsto (fun n ↦ cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)

The tower encodings converge to  $0 = \perp$  in  $\mathbb{Z}_2$ .

Proof: each stage  $\text{cnfToZp2}(\text{towerNONote } (n+1)) = 2^{n+1}$  in  $\mathbb{Z}_2$ , so its 2-adic norm is  $\|2\|^{n+1} = (1/2)^{n+1} \rightarrow 0$ . Uses Metric.tendsto\_atTop and exists\_pow\_lt\_of\_lt\_one.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Section V: Ordinal Tower Limit and Snap Threshold

This section connects the ordinal tower to the ZPE machine-phase snap. The cofinality theorem shows that the fundamental sequence approaches  $\varepsilon_0$  from below. The lower-bound theorem shows that any monotone tower-aligned map must assign  $c_0$  to all pre- $\varepsilon_0$  ordinals. A canonical witness snapping exactly at  $\varepsilon_0$  is exhibited.

### What this does NOT claim (§V)

Gentzen's theorem: that  $\varepsilon_0$  is the proof-theoretic ordinal of PA (not claimed — see Remark R-L.1).

Any statement about formal provability in PA.

That  $\varepsilon_0$  is the UNIQUE minimal snap boundary: snap\_threshold\_is\_epsilon\_zero shows no ordinal below  $\varepsilon_0$  works (for maps satisfying the stated hypotheses), but does not rule out maps that snap at some ordinal strictly above  $\varepsilon_0$ .

That the snap threshold result applies to all maps  $\text{Ordinal} \rightarrow \text{MachinePhase}$  regardless of the monotonicity and tower-alignment hypotheses.

### Theorem: fundamentalSeq\_cofinal

$\forall \alpha : \text{Ordinal}, \alpha < \varepsilon_0 \rightarrow \exists n : \mathbb{N}, \alpha < \text{fundamentalSeq } n$

The fundamental sequence is cofinal in  $\varepsilon_0$ : for any ordinal below  $\varepsilon_0$ , some tower stage exceeds it.

Proof:  $\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$  (epsilonZero\_eq\_nfp), and  $\text{lt\_nfp\_iff}$  gives  $a < \text{nfp } f b \leftrightarrow \exists n, a < f^n[n] b$ .

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: snap\_threshold\_is\_epsilon\_zero

For any  $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$  satisfying:

### Theorem: snap\_threshold\_is\_epsilon\_zero

(a) hmono:  $\forall \alpha \leq \beta, \text{join}(\phi \alpha)(\phi \beta) = \phi \beta$  (order-non-decreasing in the ZPSemilattice sense:  $\phi \alpha \leq \phi \beta$ )

(b) h0:  $\forall n : \mathbb{N}, \phi(\text{fundamentalSeq } n) = c_0$

we have:  $\forall \alpha < \varepsilon_0, \phi \alpha = c_0$

This is a lower bound: no ordinal below  $\varepsilon_0$  is a snap point for maps satisfying (a) and (b). "Minimal" not "unique."

Proof: cofinality gives a stage above  $\alpha$ ; monotonicity chains  $\phi \alpha \leq \phi(\text{stage}) = c_0$ ;  $c_0 = \perp$  forces  $\phi \alpha = c_0$ .

Lean purity: [propxt, Classical.choice, Quot.sound]. ✓

### Theorem: c1\_epsilon\_zero\_identification

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi(\text{fundamentalSeq } n) = c_0) \wedge \phi \varepsilon_0 = c_1$

Witness:  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1.$

One specific map sends all tower stages to  $c_0$  and  $\varepsilon_0$  to  $c_1$ .

The stronger structural claim — an order-preserving morphism  $\text{Ordinal} \rightarrow_o \text{MachinePhase}$  compatible with the CNF  $\rightarrow \mathbb{Z}_2$  encoding — remains outside Lean scope: no type bridge between Ordinal and MachinePhase is defined in this library.

Lean purity: [propxt, Classical.choice, Quot.sound]. ✓

## Section VI: Kleene-Ordinal Fixed-Point Bridge

The ordinal fixed-point structure ( $\varepsilon_0 = \text{nfp}(\omega^\wedge \cdot) 0, \omega^\wedge \varepsilon_0 = \varepsilon_0$ ) and the computational fixed-point structure (Kleene's recursion theorem, `roger_fixed_point_stability`) both require `Classical.choice` at their non-constructive step — parallel structure, not a proved isomorphism. The hypothesis `hfp` encodes that ordinal fixed points of  $\omega^\wedge \cdot$  map to the snap state  $c_1$ . Under this hypothesis plus monotonicity and tower alignment,  $\varepsilon_0$  is the minimal snap threshold.

### Theorem: snap\_exactly\_at\_epsilon\_zero

For any  $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$  satisfying:

(a) hmono: order-non-decreasing ( $\text{join}(\phi \alpha)(\phi \beta) = \phi \beta$  for  $\alpha \leq \beta$ )

(b) h0: every tower stage maps to  $c_0$

(c) hfp: every fixed point of  $\omega^\wedge \cdot$  maps to  $c_1$

we have:  $\phi \varepsilon_0 = c_1$  AND  $\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

### Theorem: snap\_exactly\_at\_epsilon\_zero

$\varepsilon_0$  is the minimal snap threshold:  $\phi$  assigns  $c_1$  first at  $\varepsilon_0$ .

Note: "minimal" not "unique." A map satisfying these hypotheses could also assign  $c_1$  to ordinals above  $\varepsilon_0$ ; what is ruled out is any snap strictly before  $\varepsilon_0$ .

Proof: upper bound from hfp + epsilonZero\_fixedPoint; lower bound from snap\_threshold\_is\_epsilon\_zero.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: kleene\_ordinal\_snap\_bridge

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge$

$(\forall \alpha, \omega^\alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge$

$\phi \varepsilon_0 = c_1 \wedge$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

Witness:  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ .

Note: the theorem name refers to the informal §VI conceptual parallel between Kleene recursion fixed points and ordinal fixed points of  $\omega^\cdot$ . The theorem itself is purely ordinal — no Code or eval appears.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Section VII: Canonical Snap Map — Full Closure

The canonical threshold map  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$  satisfies all three conditions of snap\_exactly\_at\_epsilon\_zero (hmono, h0, hfp). For this specific map, the snap identification is unconditional: all five conditions are simultaneously verified without free hypotheses.

### Theorem: snap\_map\_mono

$\forall \alpha \beta : \text{Ordinal}, \alpha \leq \beta \rightarrow$

$\text{join } (\text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1)$

$(\text{if } \beta < \varepsilon_0 \text{ then } c_0 \text{ else } c_1) =$

$\text{if } \beta < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$

The canonical map is order-non-decreasing.

Proof: four cases ( $\alpha/\beta$  vs  $\varepsilon_0$ ). The case  $\alpha \geq \varepsilon_0$  with  $\beta < \varepsilon_0$  is impossible by  $\alpha \leq \beta$ . Each live case closes by rfl from the MachinePhase join definition.



### Theorem: snap\_map\_mono

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: epsilon\_zero\_snap\_canonical

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall \alpha \beta, \alpha \leq \beta \rightarrow \text{join } (\phi \alpha) (\phi \beta) = \phi \beta) \wedge [\text{hmono}]$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge [\text{h0}]$

$(\forall \alpha, \omega^\alpha \alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge [\text{hfp}]$

$\phi \varepsilon_0 = c_1 \wedge [\text{snap}]$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$  [minimality]

All five conditions verified for the explicit witness  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ , with no free hypotheses.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: snap\_zp2\_correspondence

Four independent facts about the same tower sequence:

(i)  $\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

(ii)  $\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0$  where  $\phi$  is the canonical map

(iii)  $\text{Filter.Tendsto } (\text{fun } n \mapsto \text{cnfToZp2 } (\text{towerNONote } n)) \text{ Filter.atTop } (\text{nhds } 0)$

(iv)  $\phi \varepsilon_0 = c_1$

The same tower sequence witnesses both the ordinal approach to  $\varepsilon_0$  and the 2-adic approach to 0. The full structural identification ( $\varepsilon_0 \leftrightarrow \perp$  via a type bridge) remains outside Lean scope — see §V.

Proof:  $\langle \text{epsilonZero\_tower\_lt}, \text{fun } n \mapsto \text{if\_pos } \dots, \text{tower\_converges\_to\_zero}, \text{if\_neg } (\text{lt\_irrefl } \varepsilon_0) \rangle$

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Remaining Formal Gap

### The Remaining Gap

The identification of ZPE's MachinePhase element  $c_1$  with the ordinal  $\varepsilon_0$  via a formal type bridge remains outside Lean scope.

## The Remaining Gap

What is proved: a canonical map  $\text{Ordinal} \rightarrow \text{MachinePhase}$  assigns  $c_1$  exactly at  $\epsilon_0$  and nowhere earlier. What is not proved: a canonical ZPSemilattice morphism  $\text{MachinePhase} \rightarrow \mathbb{Z}_2$  that would connect ZPE's  $\perp = c_0$  to ZPB's  $\perp = 0$  formally.

Such a bridge would require:

- (1)  $\text{snapEmbed} : \text{MachinePhase} \rightarrow \mathbb{Z}_2$  mapping  $c_0 \mapsto 0, c_1 \mapsto 1$
- (2) proof that  $\text{snapEmbed}$  is join-preserving
- (3) a bridge theorem deriving  $\text{hfp}$  from  $\text{tower\_converges\_to\_zero}$  via  $\text{snapEmbed}$

This would make  $\text{hfp}$  a theorem rather than a hypothesis in  $\text{snap\_exactly\_at\_epsilon\_zero}$ . The canonical witness ( $\text{epsilon\_zero\_snap\_canonical}$ ) satisfies all five conditions without this bridge; the bridge would close the gap between the two formal instances of  $\perp$  across ZPE and ZPB.

## Theorem Summary

| Theorem                                      | Section | Status   |
|----------------------------------------------|---------|----------|
| <code>roger_fixed_point_stability</code>     | §II     | Proved ✓ |
| <code>epsilonZero_fixedPoint</code>          | §III    | Proved ✓ |
| <code>epsilonZero_eq_nfp</code>              | §III    | Proved ✓ |
| <code>epsilonZero_eq_iSup</code>             | §III    | Proved ✓ |
| <code>epsilonZero_tower_lt</code>            | §III    | Proved ✓ |
| <code>epsilonZero_le_fixedPoint</code>       | §III    | Proved ✓ |
| <code>fundamentalSeq_strictMono</code>       | §III    | Proved ✓ |
| <code>tower_stage_zero / _one / _two</code>  | §III    | Proved ✓ |
| <code>towerNONote_repr</code>                | §IV     | Proved ✓ |
| <code>cnfToZp2_tower_valuation</code>        | §IV     | Proved ✓ |
| <code>cnfToZp2_valuation_unbounded</code>    | §IV     | Proved ✓ |
| <code>fundamentalSeq_zp2_converges</code>    | §IV     | Proved ✓ |
| <code>tower_converges_to_zero</code>         | §IV     | Proved ✓ |
| <code>zpe_snap_ordinal_correspondence</code> | §V      | Proved ✓ |
| <code>epsilonZero_tower_bound</code>         | §V      | Proved ✓ |
| <code>c1_epsilon_zero_identification</code>  | §V      | Proved ✓ |

| Theorem                        | Section | Status   |
|--------------------------------|---------|----------|
| fundamentalSeq_cofinal         | §V      | Proved ✓ |
| snap_threshold_is_epsilon_zero | §V      | Proved ✓ |
| snap_exactly_at_epsilon_zero   | §VI     | Proved ✓ |
| kleene_ordinal_snap_bridge     | §VI     | Proved ✓ |
| snap_map_mono                  | §VII    | Proved ✓ |
| epsilon_zero_snap_canonical    | §VII    | Proved ✓ |
| snap_zp2_correspondence        | §VII    | Proved ✓ |

### Axiom Purity

All 23 theorems carry axiom footprint: [propext, Classical.choice, Quot.sound].

These are standard Mathlib infrastructure axioms (ordinal theory, p-adic analysis, computability). They are not ZP-L commitments.

Classical.choice is load-bearing: it is the formal non-constructivity appearing at the diagonal step in each ZP layer (§I). Its presence is expected and documented, not incidental.

Zero sorry in ZPL.lean. Verified: lake build, May 2026.